

Statistical Machine Learning

Activation Functions

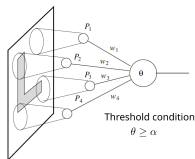
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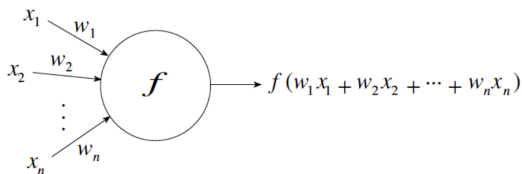
What are activation functions?

Recall that in the threshold logic, once we receive the input, a decision must be made to *fire* or *suppress* the output.



Activation functions

Activation functions generalize the concept of activation given a specific input.

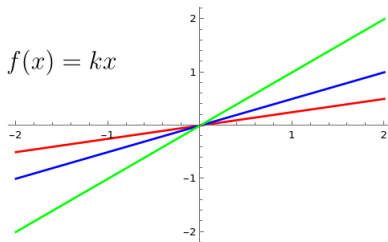


Linear Functions

These functions are used to model the firing rate of a neuron.

$$f(x) = kx \quad k \text{ is a positive constant}$$

Pros: It is continuous and differentiable.

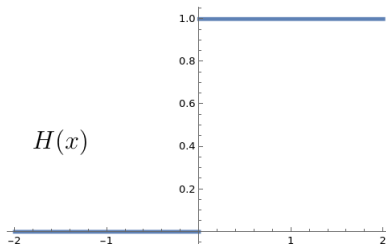


Step Functions

These are biologically inspired type of activation.

- ① Heaviside function (threshold step function). This function fires only for positive values.

$$H(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 & \text{if } x > 0 \end{cases}$$



Step Functions (cont)

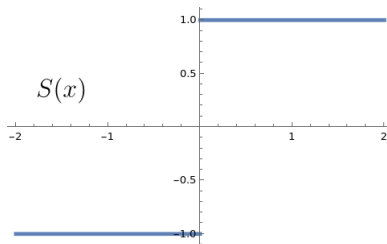
This function is differentiable except in $x = 0$. Informally, people refers to the Dirac delta function as its *derivative*.

$$H'(x) = \delta(x) = \begin{cases} 0 & \text{if } x \neq 0, \\ \infty & \text{if } x = 0 \end{cases} \quad \text{it also satisfies } \int_{-\infty}^{\infty} \delta(x) dx = 1$$

Step Functions (cont)

- ② Signum Function. This is a variation of the Heaviside function

$$S(x) = \begin{cases} -1 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$



Hockey stick functions

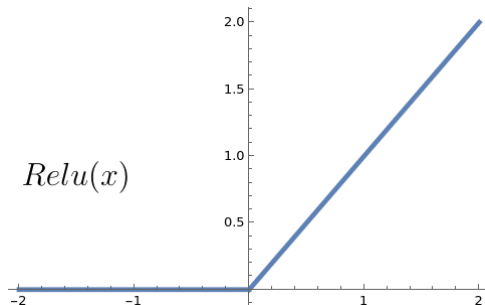
- 1 Rectified Linear Unit (ReLU). The activation is linear for $x \geq 0$.

$$\text{ReLU}(x) = xH(x) = \max\{x, 0\} = \begin{cases} 0 & \text{if } x < 0, \\ x & \text{if } x \geq 0. \end{cases}$$

Pros:

- This function does not saturate (see below with sigmoid functions)
- Neural networks with ReLU activation functions tend to learn several times faster than similar networks with saturating activation function.

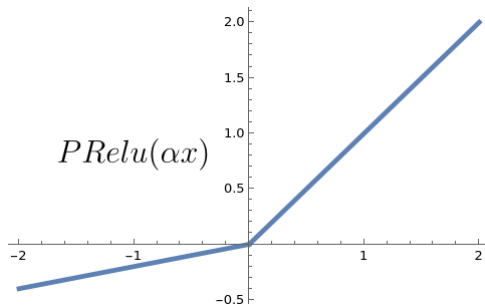
Hockey stick functions



- ② Parametric Rectified Linear Unit (PReLU). In this case the activation is piecewise linear, having different firing rates for $x < 0$ and $x > 0$

$$PReLU(\alpha, x) = \begin{cases} \alpha x & \text{if } x < 0, \\ x & \text{if } x \geq 0. \end{cases} \quad \alpha > 0.$$

Hockey stick functions (cont)



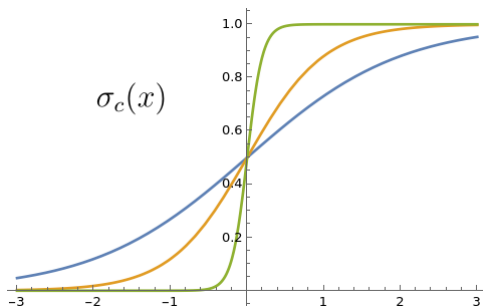
Sigmoid functions

These types of activation functions have that advantage that they are smooth and can approximate a step function to any degree of accuracy.

- 1 Logistic function with parameter $c > 0$.

$$\sigma_c(x) = \frac{1}{1 + \exp(-cx)}$$

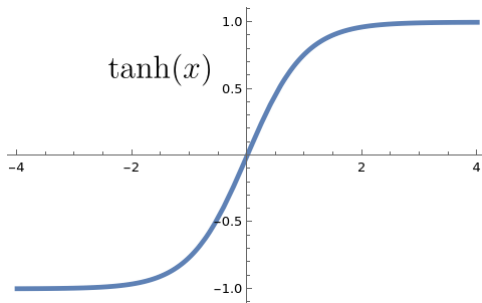
The parameter $c > 0$ controls the firing rate of the neuron. Large values of c correspond to a fast change of values from 0 to 1.



Sigmoid Functions

1 Hyperbolic tangent.

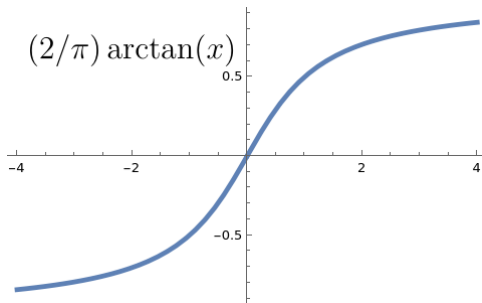
$$\tanh(x) = \frac{\exp(x) - \exp(-x)}{\exp(x) + \exp(-x)}$$



Sigmoid Functions

- 1 Arctangent function.

$$h(x) = \frac{2}{\pi} \arctan(x)$$



Summary of Activation Functions

We have seen various types of activation function. All these functions have some characteristic in common, more specifically

- they are well defined for every point in the real line,
- they are non-decreasing,
- they are smooth almost everywhere.

Cost Functions

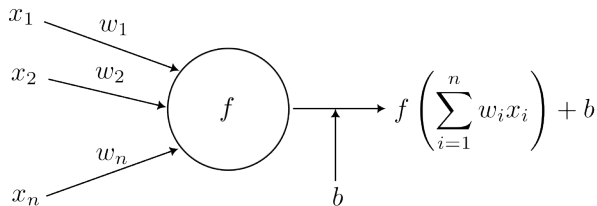
In the learning process, the parameters of a neural network are subject to minimize a certain objective function which represents a measure of proximity between the prediction of the network and the associated target. Note that these functions are also called *error functions* or *loss functions*. For instance, in multi linear regression we have RSS as the cost function.

Input, Output and Target

The **input** of a neural networks is obtained from given data or from sensors that perceive the environment. The input is a variable that is fed into the network. It can be one-dimensional variable x , or a vector $\mathbf{x} \in \mathbb{R}^n$, a matrix, a tensor, or a random variable X .

The network acts as a function and provides an **output** which can be one-dimensional $y \in \mathbb{R}$, or a vector $\mathbf{y} \in \mathbb{R}^n$, a matrix, a tensor, or a random variable Y . We normally denote the **input-output** mapping by $f_{w,b}$. With the previous notations $f_{w,b}(x) = y$, $f_{w,b}(\mathbf{x}) = \mathbf{y}$, and $f_{w,b}(X) = Y$.

Input, Output, and Target (cont)



The **target function** is the desired functional relationship that the network tries to approximate. This function is independent of the parameters w and will be denoted by $z = \phi(x)$, $\mathbf{z} = \phi(\mathbf{x})$ or $Z = \phi(X)$.

Goal

The neural network will tune the parameters (w, b) until the output variable y will be in a proximity of the target variable z .

The proximity will be determined by the cost function

$$C(w, b) = \text{distance}(y, z)$$

The optimal parameters of the network are given by

$$(w^*, b^*) = \arg \min_{w, b} C(w, b)$$

The process by which the parameters (w, b) are tuned into (w^*, b^*) is called **learning**.

Supremum Cost Function

Let's assume that a neural network takes inputs in $x \in [0, 1]$, is supposed to learn a given continuous function $\phi: [0, 1] \rightarrow \mathbb{R}$. If $f_{w,b}$ is the input-output mapping of the network the associated cost function is

$$C(w, b) = \sup_{x \in [0, 1]} |f_{w,b}(x) - \phi(x)|$$

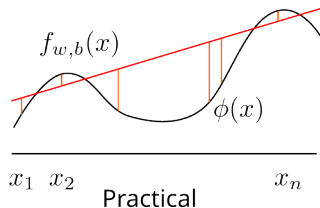
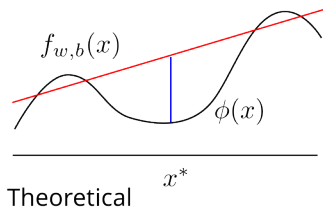
For all practical purposes, when the target function is known at n points

$$z_1 = \phi(x_1), z_2 = \phi(x_2), \dots, z_n = \phi(x_n)$$

then, the supremum cost function becomes

$$C(w, b) = \max_{1 \leq i \leq n} |f_{w,b}(x_i) - z_i|$$

Sup Cost Function (cont)



L^2 Cost Function

Assume the input of the network is $x \in [0, 1]$ and that the target function $\phi: [0, 1] \rightarrow \mathbb{R}$ is square integrable. If $f_{w,b}$ is the input-output mapping, the associated cost function measures the distance in the L^2 norm between the output and the target

$$C(w, b) = \int_0^1 (f_{w,b}(x) - \phi(x))^2 dx.$$

If the target function is known at only n points

$$z_1 = \phi(x_1), z_2 = \phi(x_2), \dots, z_n = \phi(x_n)$$

then this cost function becomes the square of the Euclidean distance in $\mathbb{R}^n$ between $f_{w,b}$ and $\mathbf{z}$

$$C(w, b) = \sum_{i=1}^n (f_{w,b}(x_i) - z_i)^2 = \|f_{w,b}(\mathbf{x}) - \mathbf{z}\|^2$$

Mean Square Error Cost Function

Consider a neural network whose input is a random variable X , and its output is the random variable $Y = f_{w,b}(X)$. Assume that the network is used to approximate the target random variable Z . The cost function will measure the proximity between the output and the target random variables Y and Z . A good candidate is given by the expectation of their squared difference

$$C(w, b) = \mathbb{E}[(Y - Z)^2] = \mathbb{E}[(f_{w,b}(X) - Z)^2]$$

Let's consider the so-called **training set** consisting of n measurements of random variables (X, Z) , which are given by (x_i, z_i) . Then the cost function becomes the **empirical mean of the square difference of Y and Z** .

$$\hat{C}(w, b) = \frac{1}{n} \sum_{j=1}^n (f_{w,b}(x_j) - z_j)^2$$

Cross-entropy

Let p and q be two densities in $\mathbb{R}$ (loosely speaking, this means that these functions can describe a random variable).

The negative likelihood function $-\ln(q(x))$ measures the information given by $q(x)$. The **cross-entropy** of p with respect to q is defined as

$$S(p, q) = - \int_{-\infty}^{\infty} p(x) \ln q(x) dx$$

This represents the information given by $q(x)$ assessed from the point of view of the distribution $p(x)$.

Cross-entropy (cont)

For example let $p(x) = \exp(-x^2/2)/\sqrt{2\pi}$ be the density of the $N(0, 1)$, and $q(x) = \exp(-x^2/50)/(5 * \sqrt{2\pi})$ be the corresponding density for $N(0, 5)$.

$$S(p, q) = 13.4189 \text{ whereas } S(q, p) = 2.54838$$

The **Shannon entropy** is defined by

$$H(p) = - \int_{-\infty}^{\infty} p(x) \ln p(x) dx$$

With the same notation we have

$$H(p) = 1.41894 \text{ whereas } H(q) = 3.02838$$

Kullback-Leibler Divergence

The difference between the cross entropy and the Shannon entropy is the **Kullback-Leibler divergence**

$$D_{KL}(p, q) = S(p, q) - H(p) = \int_{-\infty}^{\infty} p(x) \ln \frac{q(x)}{p(x)} dx$$

Note that KL-divergence is not a distance, but both cross-entropy and KL-divergence can be used as cost functions for neural networks.

References

Materials and some of the pictures are from (Calin, 2019).

 Calin, O. (2019). *Deep Learning Architectures*. Springer Series in the Data Sciences. Springer. ISBN: 978-3-030-36723-7.

I have used some of the graphs by hacking TiKz code from StakExchange, Inkscape for more aesthetic plots and other old tricks of $\text{\TeX}$. The entropy calculations were done in Mathematica.